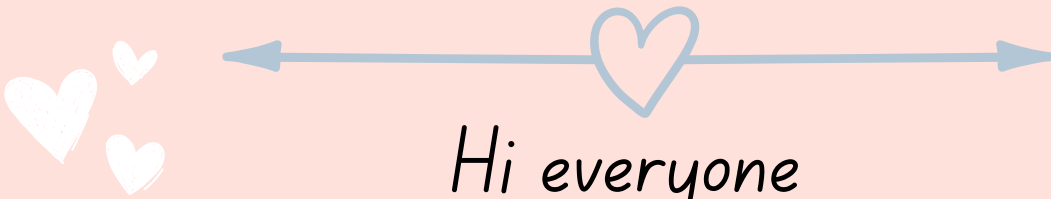


# Biology- II ADV

## LESSON-1 U-2

### Infectious Diseases

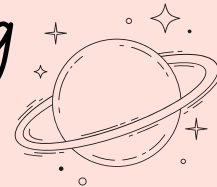


Hi everyone

I made this file to help you in your study

Hope its helpful!

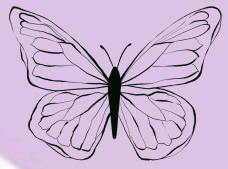
Thanks for reading



Done By:



# Pathogens Cause Infectious Disease



Infectious disease is a disease that is caused by a pathogen passed from one organism to another

pathogens are the cause of infectious diseases. Pathogens include:

- some but not all types of bacteria,
- viruses,
- protozoans,
- fungi,
- parasites.



## Germ Theory and Koch's Experiments

states that: " Some microorganisms are pathogens. "

1. Koch isolated bacteria from the blood of cattle that had died from anthrax.
2. After growing the bacteria in the laboratory, Koch injected the bacteria into healthy cattle.
3. He then isolated bacteria from the blood of newly infected cattle and grew the bacteria in the laboratory.

He analyze that:

The characteristics of the two sets of cultures were identical, indicating that the same type of bacteria caused the illness in both sets of cattle.



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# Koch's Postulates

which are rules for demonstrating that an organism causes a disease.

- **Postulate 1** The suspected pathogen must be isolated from the diseased host in every case of the disease.
- **Postulate 2** The suspected pathogen must be grown in pure culture.
- **Postulate 3** The suspected pathogen from the pure culture must cause the disease when placed in a healthy new host.
- **Postulate 4** The suspected pathogen must be isolated from the new host, grown again in pure culture, and shown to have the same characteristics as the original pathogen

| Table 1        |           | Human Infectious Diseases   |                                             |
|----------------|-----------|-----------------------------|---------------------------------------------|
| Disease        | Cause     | Affected Organ System       | How Disease is Spread                       |
| Tetanus        | Bacterium | Nervous system              | Soil in deep puncture wound                 |
| Strep throat   | Bacterium | Respiratory system          | Droplets/direct contact                     |
| Tuberculosis   | Bacterium | Respiratory system          | Droplets                                    |
| Lyme disease   | Bacterium | Skeletal and nervous system | Vector (tick)                               |
| Chicken pox    | Virus     | Skin                        | Droplets/direct contact                     |
| Rabies         | Virus     | Nervous system              | Animal bite                                 |
| Common cold    | Virus     | Respiratory system          | Droplets/direct contact                     |
| Influenza      | Virus     | Respiratory system          | Droplets/direct contact                     |
| Hepatitis B    | Virus     | Liver                       | Direct contact with exchange of body fluids |
| West Nile      | Virus     | Nervous system              | Vector (mosquito)                           |
| Giardia        | Protozoan | Digestive tract             | Contaminated water                          |
| Malaria        | Protozoan | Blood and liver             | Vector (mosquito)                           |
| Athlete's foot | Fungus    | Skin                        | Direct contact or contaminated objects      |

A pure culture is a culture that contains no other types of microorganisms—only the suspected pathogen.

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## reservoir

### Human Reservoirs

the main reservoir  
They might pass the pathogen directly or indirectly to other humans.

### Animal Reservoirs

EX. Influenza and rabies  
Rabies is found in many wild animals

### Other Reservoirs

#### Examples:

Water

Food

Soil

## Symptoms of Disease

the cells are damaged, they might die, causing symptoms in the host.

Some disease symptoms, such as coughing and sneezing, are triggered by the immune system.

## Transmission of Pathogens

### • Direct Contact :

Touching

### • Indirect Contact:

Air

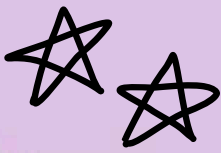
### • Vectors:

Mosquitoes

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# Disease Patterns



**Endemic** diseases, such as the common cold, are continually found in small amounts within the population.

**Epidemic** occurs when a particular disease will have a large outbreak in an area and afflict many people.

**Pandemic** occurs when an epidemic is widespread throughout a large region, such as a country, continent, or the entire globe.



## Treating and Fighting Diseases

One type of prescription drug is an antibiotic, which is a substance that can kill or inhibit the growth of microorganisms.

Penicillin is secreted by the fungus *Penicillium*. This fungus secretes the chemical penicillin to kill competing bacteria that grow on the fungal food source.

antibiotics has caused many bacteria to become resistant to particular antibiotics.

Because reproduction can occur so rapidly in bacteria, the number of antibiotic-resistant bacteria in a population can increase quickly, too.

staphylococcal disease—it is acquired in high-density living conditions, which can result in skin infections, pneumonia, or meningitis.

# The End



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